

TYLSTRUP, Denmark

WMO: 060310

Lat: 57.185N

Lon: 9.953E

Elev: 43

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.6	(c) 19.5	(d) 10.7	(e) 9.5	(f) 16.5	(g) 15.0	(h) 11.8	(i) 22.6	(j) 43.4	(k) 36.7	(l) 39.0	(m) 34.8	(n) 3.4	(o) 10	(p) 0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	77.9	64.6	74.7	63.6	71.5	61.8	66.7	74.0	65.1	71.6	63.6	69.5	8.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	90.0	69.2	62.6	84.9	67.6	61.0	80.4	66.2	31.3	74.3	30.1	71.8	28.9	69.7	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 31.4	(b) 25.6	(c) 21.2	(e) 9.3	(f) 81.7	(g) 6.5	(h) 3.4	(i) 4.6	(j) 84.2	(k) 0.8	(l) 86.2	(m) -2.8	(n) 88.1	(o) -7.6	(p) 90.7		
(a) 31.4	(b) 25.6	(c) 21.2	(e) 9.1	(f) 69.3	(g) 6.2	(h) 1.7	(i) 4.6	(j) 70.5	(k) 1.0	(l) 71.4	(m) -2.5	(n) 72.4	(o) -7.0	(p) 73.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	33.4	33.4	37.6	45.0	51.9	57.1	62.2	61.0	55.8	48.4	41.7	36.7		
	DBStd	11.61	6.98	6.84	6.59	5.52	5.93	4.42	4.43	3.65	4.30	5.30	5.92	7.42		
	HDD50	2343	516	464	384	169	46	1	0	0	6	93	253	411		
	HDD65	6580	981	884	849	601	409	242	113	134	277	515	699	876		
	CDD50	1282	0	0	1	18	104	214	379	340	179	43	4	0		
	CDD65	42	0	0	0	0	1	4	27	9	1	0	0	0		
	CDH74	343	0	0	0	0	15	51	236	38	3	0	0	0		
	CDH80	37	0	0	0	0	0	3	33	1	0	0	0	0		
	WSAvg	8.8	10.5	10.1	9.9	8.7	8.3	8.2	7.2	7.6	7.9	8.2	9.5	10.1		
	PrecAvg	24.5	1.8	1.2	1.4	1.4	1.8	2.3	2.5	2.7	2.6	2.6	2.4	2.0		
(15) Precipitation	PrecMax	34.4	4.0	3.5	3.4	3.2	4.7	4.9	7.9	8.0	7.0	6.0	4.7	4.3		
	PrecMin	17.0	0.2	0.1	0.0	0.2	0.2	0.5	0.2	0.4	0.8	0.6	0.6	0.4		
	PrecStd	4.2	0.9	0.7	0.7	0.7	0.9	1.3	1.6	1.5	1.2	1.3	1.0	1.0		
	PrecStd	4.2	0.9	0.7	0.7	0.7	0.9	1.3	1.6	1.5	1.2	1.3	1.0	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	47.8	56.4	66.1	76.3	79.2	83.6	78.0	72.7	63.2	56.7	51.4		
		MCWB	46.8	45.1	47.9	54.1	61.7	63.0	65.4	65.6	63.1	58.3	54.5	49.7		
	2%	DB	46.1	45.6	51.5	63.0	71.6	74.7	80.0	74.6	68.7	60.1	53.6	49.1		
		MCWB	44.4	43.4	45.8	52.1	58.6	62.0	65.5	64.2	61.1	56.7	51.9	47.6		
	5%	DB	44.5	43.9	48.0	59.0	67.7	70.8	76.5	71.4	65.6	58.0	51.6	47.5		
		MCWB	42.9	42.1	44.1	50.4	58.2	60.6	64.9	62.2	59.8	55.3	49.8	46.1		
	10%	DB	42.9	42.3	45.7	54.9	63.2	66.8	72.5	68.8	63.2	56.1	49.9	46.0		
		MCWB	41.3	40.7	42.6	48.4	55.4	58.8	63.1	61.5	58.2	53.7	48.5	44.5		
	0.4%	WB	47.1	46.0	49.3	56.7	64.6	66.7	69.1	68.3	64.7	59.6	54.7	50.1		
		MCDB	47.9	47.3	54.3	63.6	72.8	74.5	78.1	74.2	69.6	62.0	56.1	51.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.8	43.9	46.7	53.5	61.3	64.2	67.4	65.8	62.4	57.7	52.1	48.0		
		MCDB	45.7	45.1	50.3	60.8	68.0	71.9	75.9	71.3	66.9	59.4	53.3	48.7		
	5%	WB	43.0	42.4	44.7	51.2	58.5	61.9	65.7	64.1	60.8	55.7	50.3	46.4		
		MCDB	44.1	43.6	46.9	57.5	65.4	68.7	73.2	68.9	64.4	57.4	51.3	47.3		
	10%	WB	41.3	41.0	43.1	48.9	56.2	59.7	64.2	62.6	59.3	54.1	48.5	44.8		
		MCDB	42.5	42.2	45.2	53.8	62.1	65.0	70.8	66.9	62.3	55.7	49.8	45.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	10.3	14.5	15.6	15.0	15.5	14.1	12.6	10.6	7.8	7.4		
		MCDBR	7.8	8.2	12.9	20.9	22.5	22.9	22.6	18.5	15.8	11.2	7.5	7.4		
		MCWBR	7.8	6.9	8.6	12.0	12.6	12.4	10.8	9.4	9.7	8.4	7.0	7.5		
		MCDBR	8.1	7.8	11.3	18.9	19.5	19.4	19.6	15.3	13.6	10.1	7.5	7.5		
	5% WB	MCWBR	8.0	7.1	8.2	11.6	11.9	12.0	10.4	9.3	9.1	8.1	7.3	7.6		
		taub	0.307	0.332	0.350	0.387	0.379	0.369	0.389	0.390	0.364	0.346	0.323	0.292		
		taud	2.228	2.290	2.327	2.279	2.357	2.406	2.372	2.376	2.438	2.405	2.290	2.198		
		Ebn at Noon	175	225	254	261	272	276	267	258	248	218	168	144		
(44) All-Sky Solar Radiation	RadAvg	120	297	749	1280	1707	1879	1728	1351	909	433	159	85			
	RadStd	25	54	67	145	151	141	164	93	89	55	29	10			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	-211	-277	N/A	N/A

Nomenclature: See separate page